

Nutrition Assistant Project

Phase 7: Documentation

Introduction

The Documentation phase provides comprehensive guidance for end users and developers. It includes user manuals, setup instructions, API documentation, deployment guidance, and references required to operate, maintain, and extend the Nutrition Assistant application.

End-User Guide

Users can register with their email, log in securely, search foods, log meals, monitor calorie and macronutrient intake, view dashboard charts, and update their personal profile including health goals and dietary preferences. The dashboard provides daily nutrition summaries and progress reports.

Developer Guide

Prerequisites include Node.js, npm, Git, MongoDB, and Visual Studio Code. Developers clone the repository, install dependencies for both the client and server, configure environment variables, and start development using **npm run dev**. Git branches follow the **main**, **dev**, and **feature/*** strategy with code reviews before merging.

API Documentation

REST APIs support authentication, user profile management, food search, meal logging, and dashboard reporting. Standard endpoints include user registration, login, profile updates, food search, meal CRUD operations, and dashboard retrieval. API responses follow consistent JSON structures with success flags, messages, and data objects.

Deployment Guide

The frontend is deployed on Vercel, while the backend is deployed on Render. Environment variables are configured securely, and MongoDB is connected as the production database. After deployment, application functionality is verified through smoke testing and API validation.

Demo Video

Project demonstration video:

<https://drive.google.com/file/d/1qSYxZaZ1YlfyhAmkOxF3N0Kg1IWGIVK6/view>

Benefits of Documentation

Comprehensive documentation improves usability, simplifies maintenance, accelerates onboarding for new developers, promotes consistent development practices, and supports future enhancements.

Conclusion

The Documentation phase completes the project by providing clear instructions for users and developers, detailed API references, deployment guidance, and demonstration resources. It ensures the Nutrition Assistant can be effectively used, maintained, and enhanced over time.